

Bugs found: 4

1. Off-by-one error in loop condition: ' $i \leq 3$ ' should be ' $i < 3$ ', causing out-of-bounds access on `arr[3]` (array size is 3, valid indices are 0–2).
2. Out-of-bounds array write: `arr[3]` is written in the last iteration, which is undefined behavior (buffer overflow).
3. Integer overflow: `INT_MAX + 1` overflows a signed int, which is undefined behavior in C.
4. Use of uninitialized variable: '`soma`' is declared but never assigned before being passed to `printf`.